CPU Scheduling Simulation Report

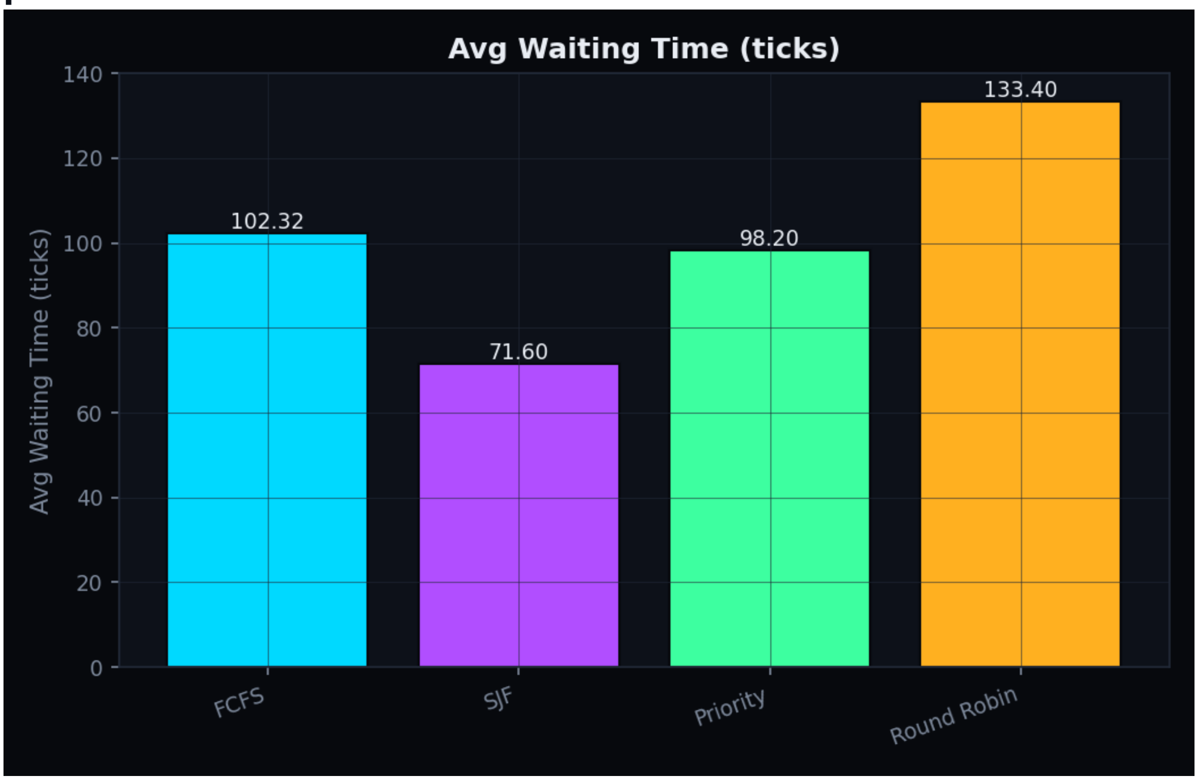
Workload: 25 processes · seed 42 · Round Robin quantum 4

Algorithm Comparison Table

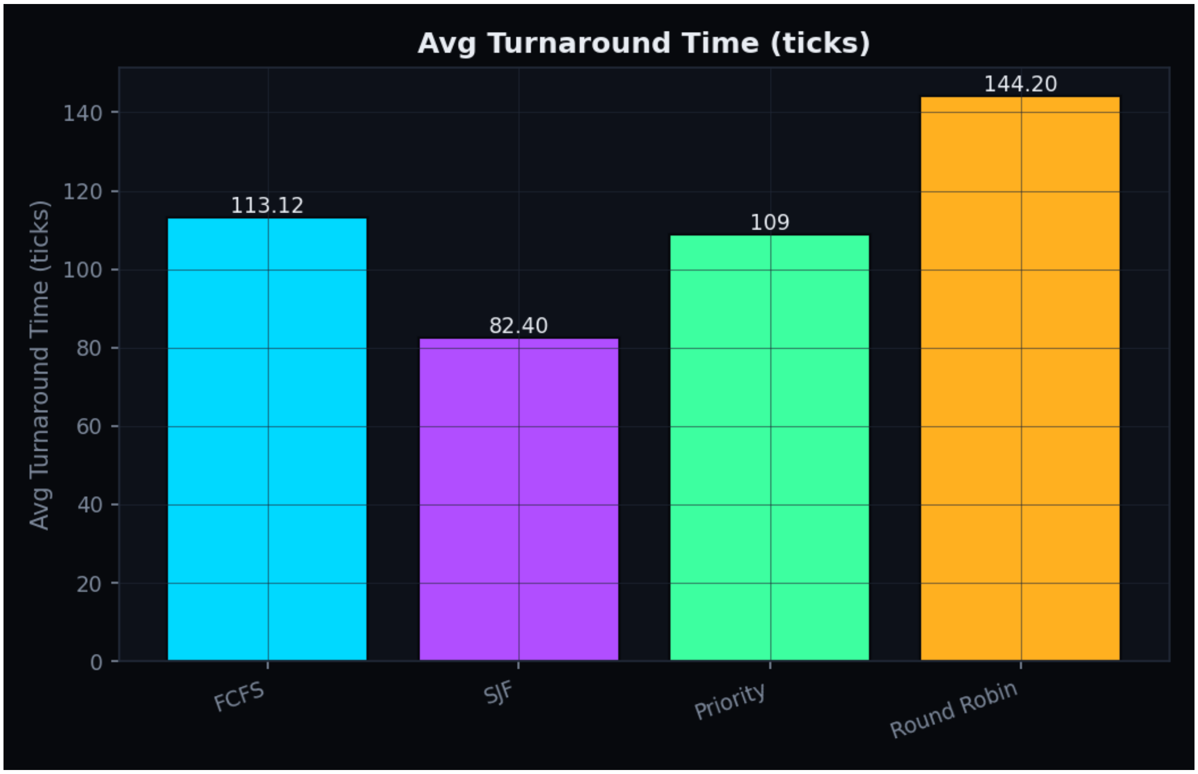
Algorithm	Avg Wait	Avg TAT	Avg Resp	CPU%	Switches	Fairness	Efficiency
FCFS	102.3	113.1	102.3	98.5	24	0.687	33.5
SJF	71.6	82.4	71.6	98.5	24	0.521	16.8
Priority	98.2	109.0	98.2	98.5	24	0.636	27.6
Round Robin	133.4	144.2	39.0	98.5	73	0.783	47.2

Fairness = Jain's Fairness Index (standard). Efficiency = a designed composite heuristic (100 × CPU utilization × fairness / (1 + starvation)), original to this project — not a standard OS metric. See README for full definitions.

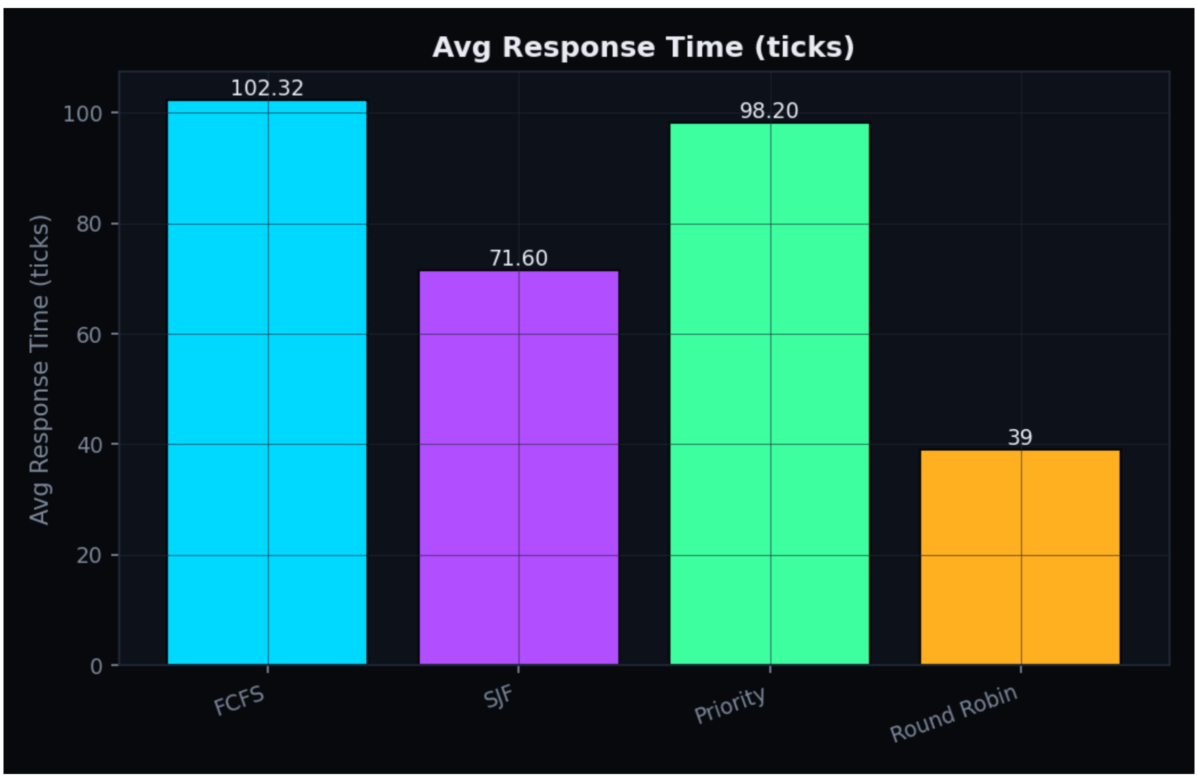
Comparison Charts



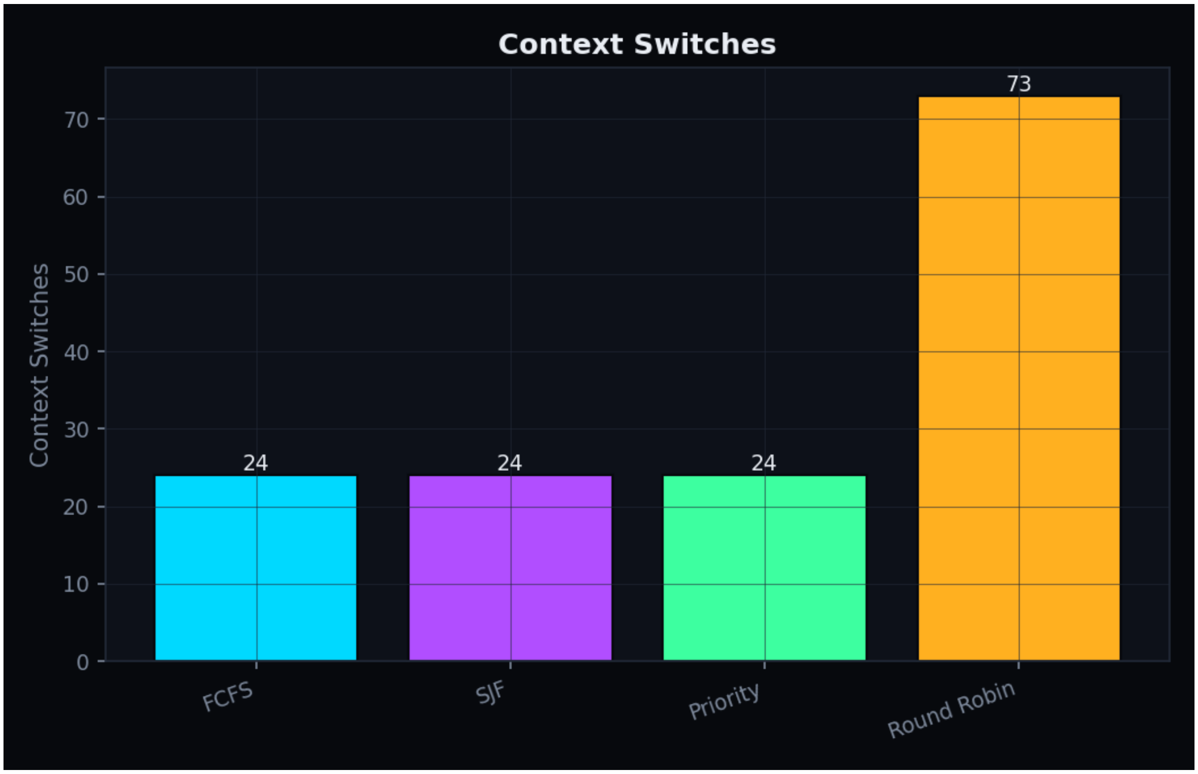
Average Waiting Time by Algorithm



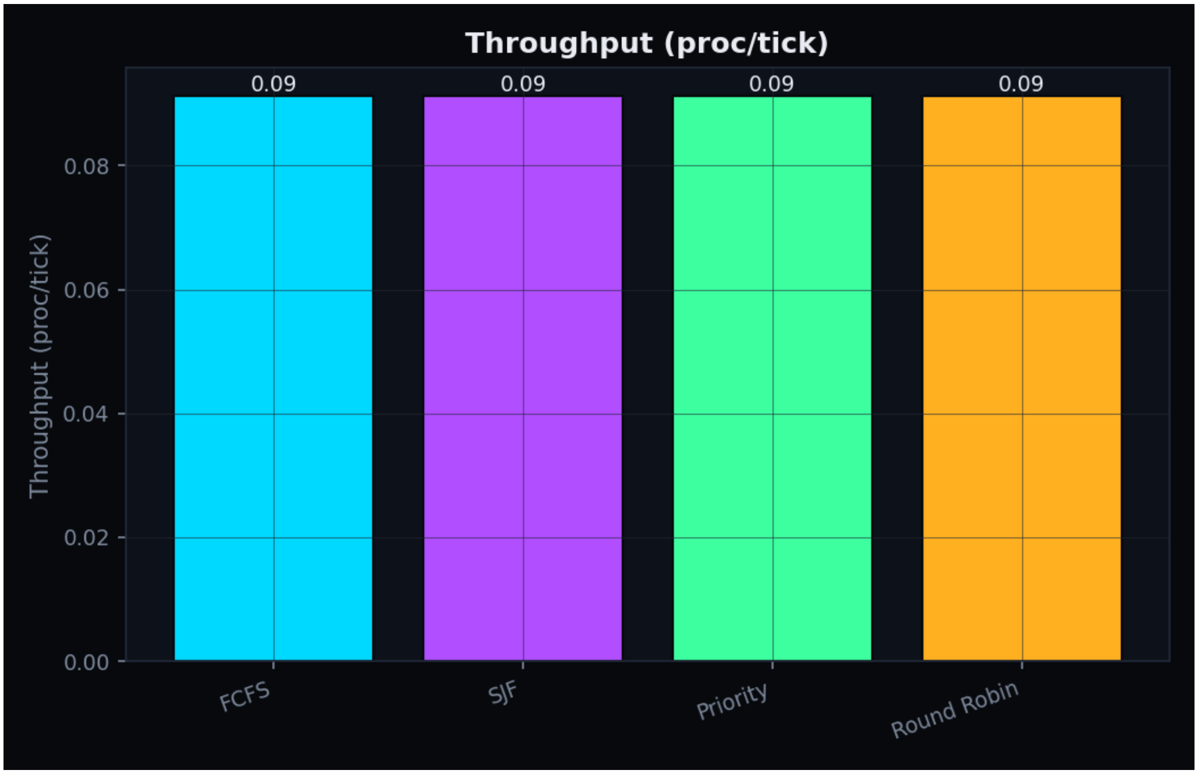
Average Turnaround Time by Algorithm



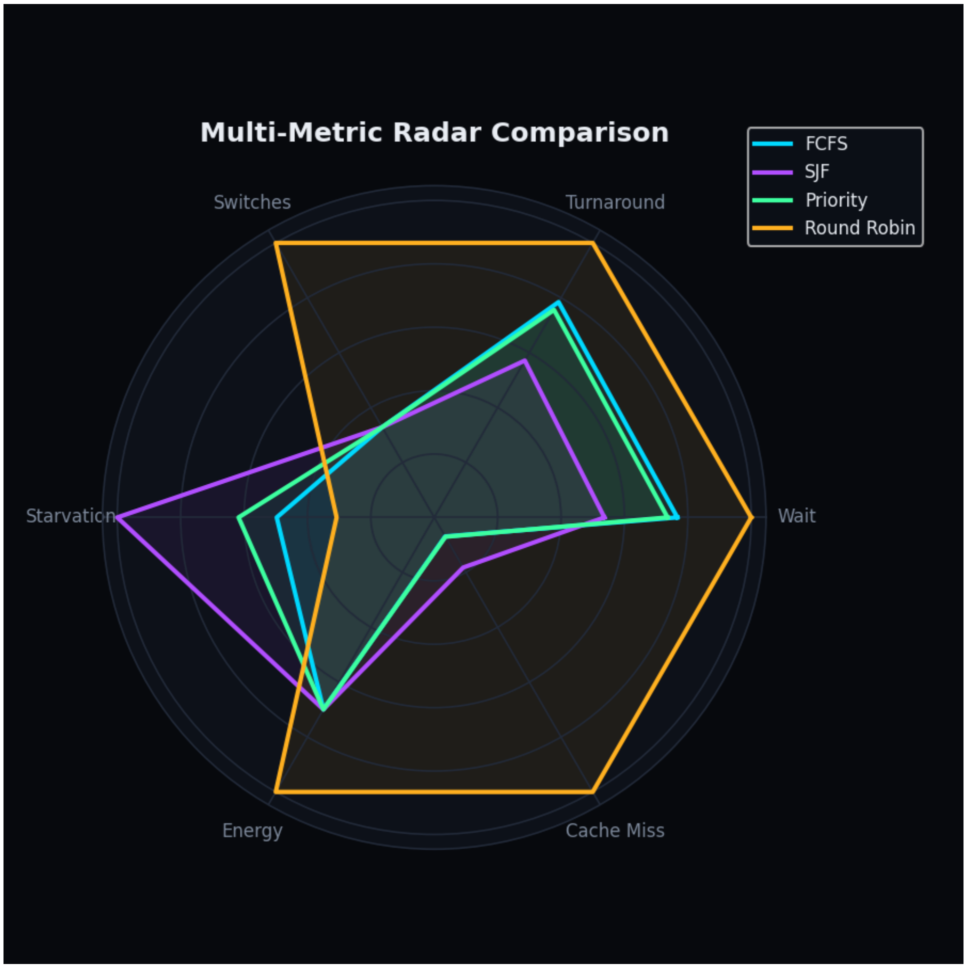
Average Response Time by Algorithm



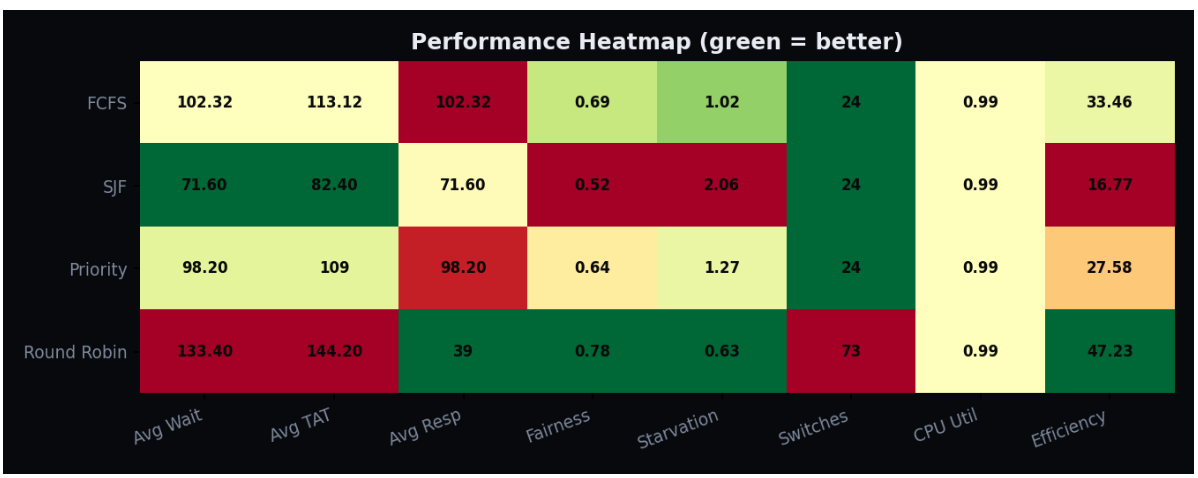
Context Switches by Algorithm



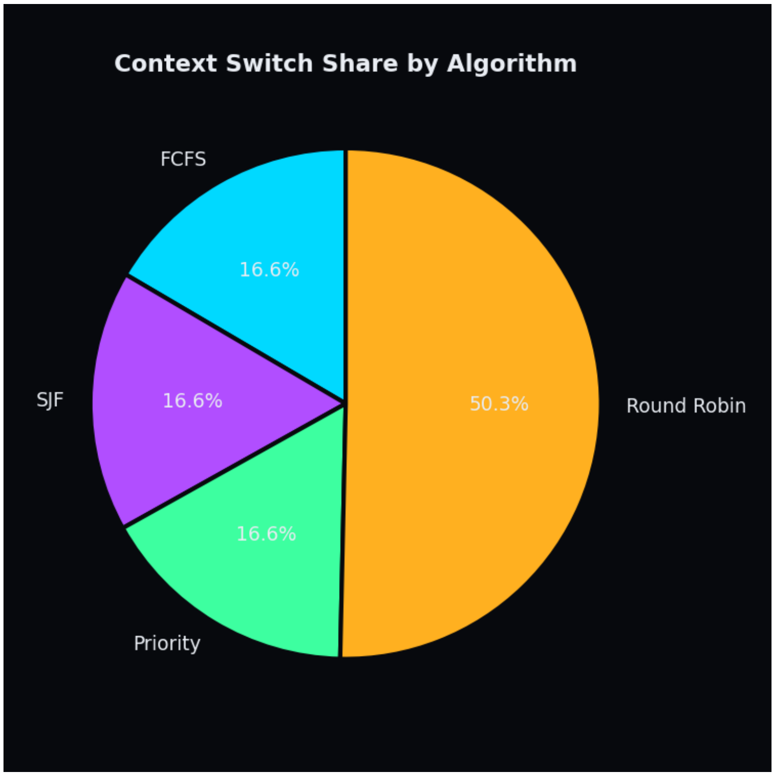
Throughput by Algorithm



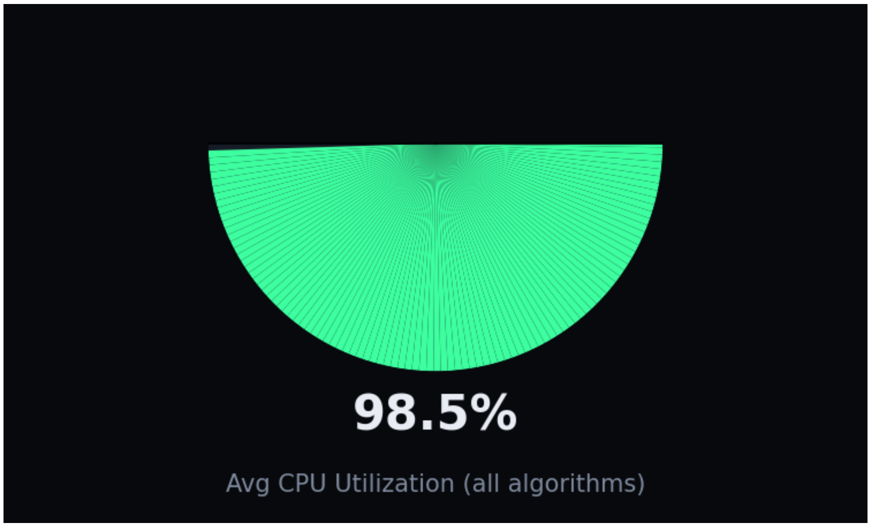
Multi-metric radar comparison (normalized per metric).



Performance heatmap — green indicates the better value per column.

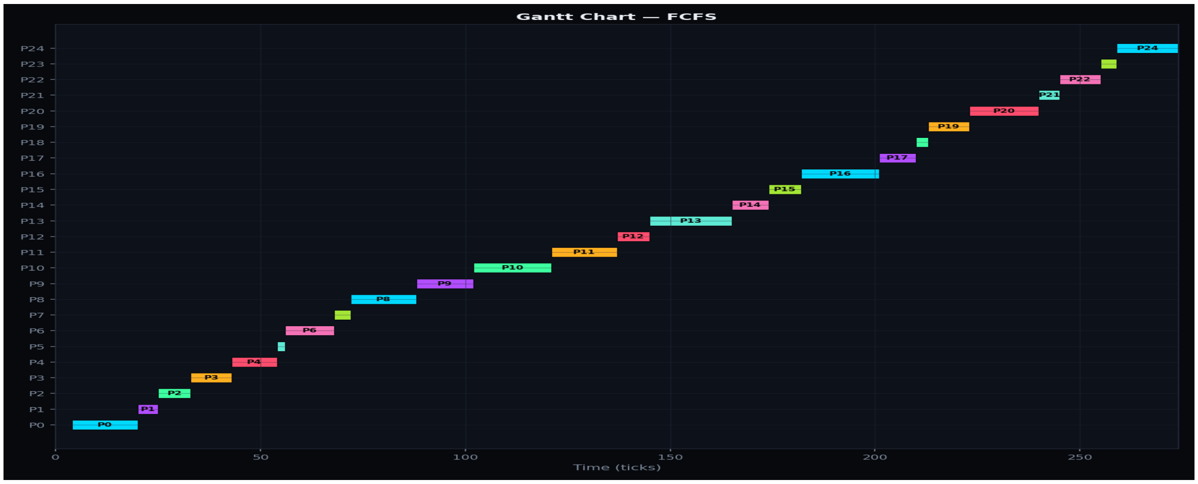


Share of total context switches contributed by each algorithm.

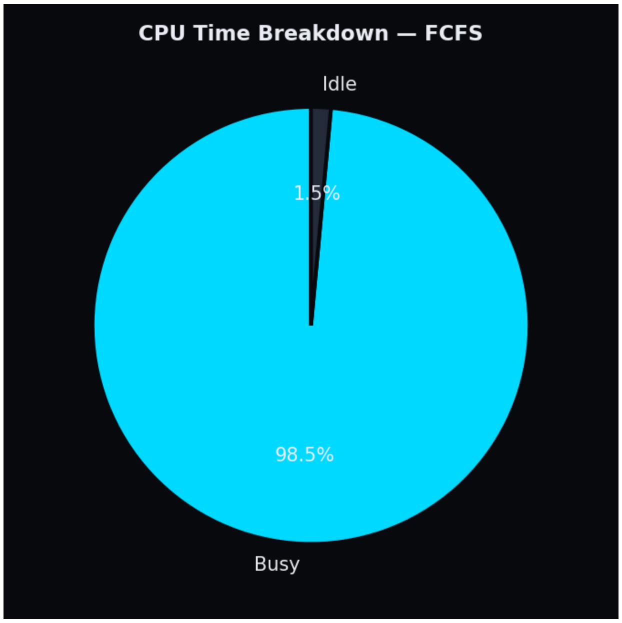


Average CPU utilization across all algorithms in this run.

Detail — FCFS



Execution timeline (Gantt) — FCFS



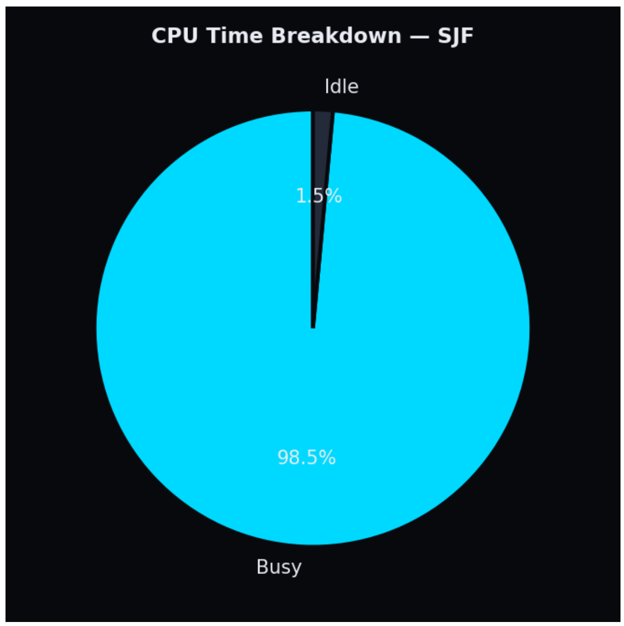
Busy vs. idle CPU time — FCFS

Metric	Value
Avg Waiting Time	102.32 ticks
Avg Turnaround Time	113.12 ticks
Avg Response Time	102.32 ticks
CPU Utilization	98.5%
Idle CPU Time	4 ticks (1.5%)
Throughput	0.0912 proc/tick
Context Switches	24
Fairness Index	0.687
Starvation Score	1.02

Scheduling Efficiency Score	33.5 / 100
Energy Score (simulated)	342 units
Cache Miss Rate (simulated)	5.9%

Detail — SJF

Execution timeline (Gantt) — SJF

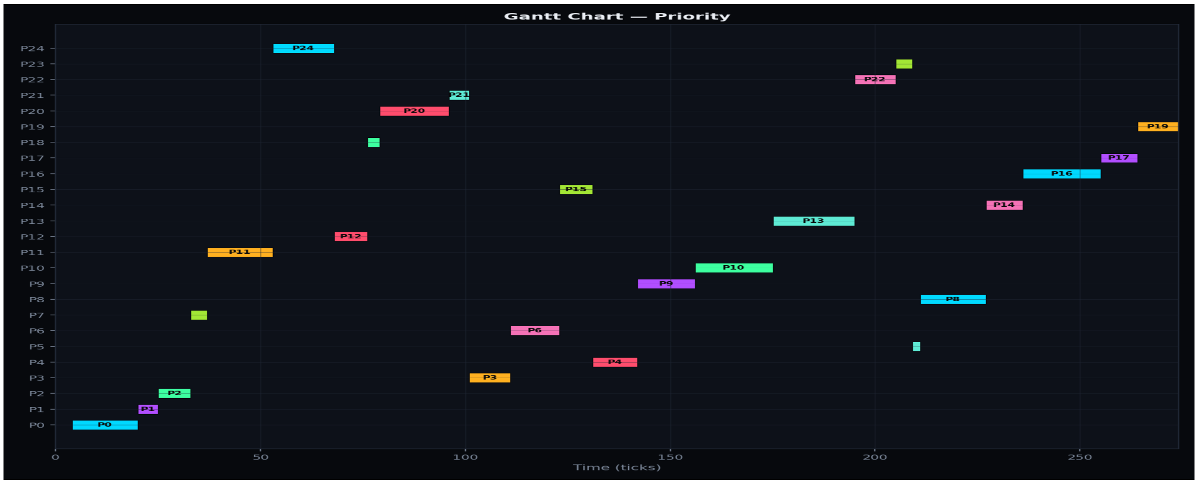


Busy vs. idle CPU time — SJF

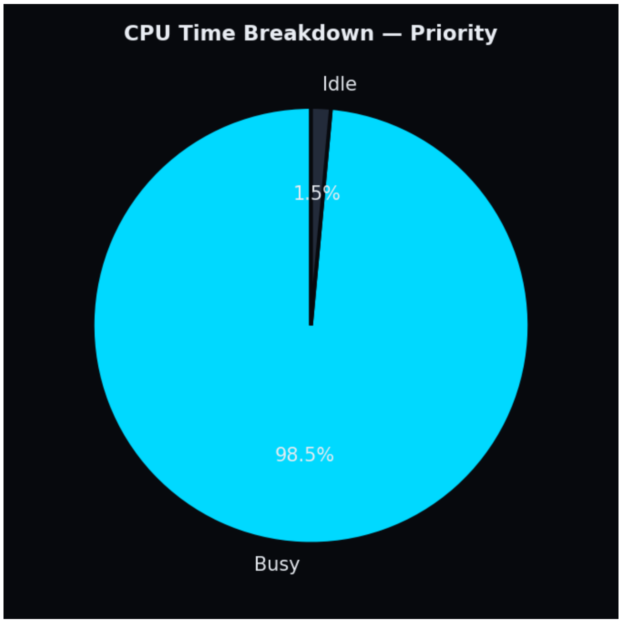
Metric	Value
Avg Waiting Time	71.60 ticks
Avg Turnaround Time	82.40 ticks
Avg Response Time	71.60 ticks
CPU Utilization	98.5%
Idle CPU Time	4 ticks (1.5%)
Throughput	0.0912 proc/tick
Context Switches	24
Fairness Index	0.521
Starvation Score	2.06

Scheduling Efficiency Score	16.8 / 100
Energy Score (simulated)	342 units
Cache Miss Rate (simulated)	15.4%

Detail — Priority



Execution timeline (Gantt) — Priority

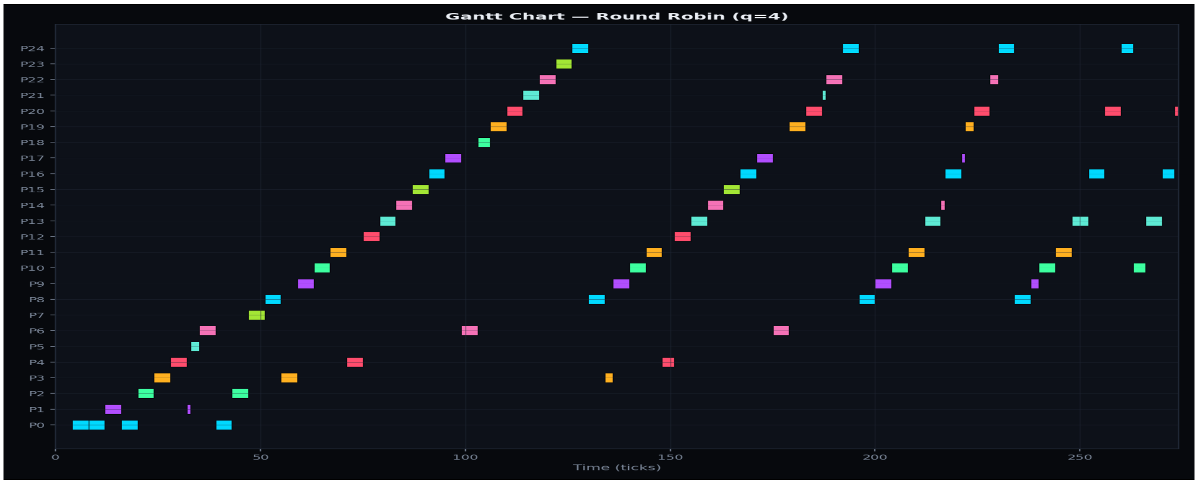


Busy vs. idle CPU time — Priority

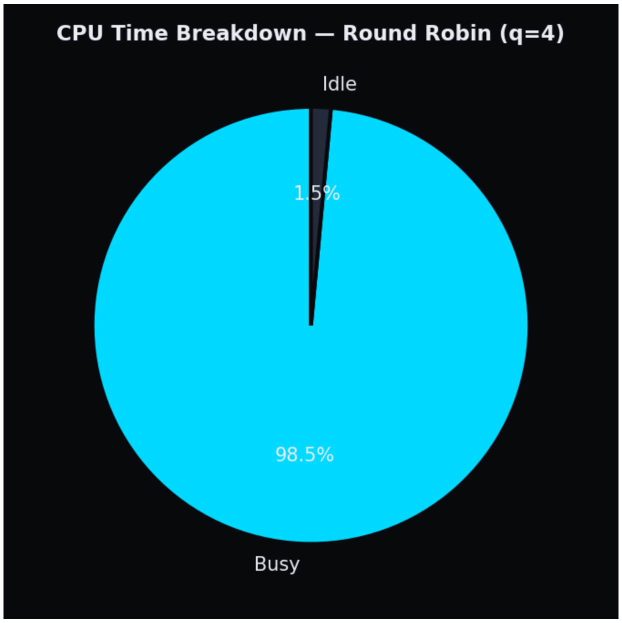
Metric	Value
Avg Waiting Time	98.20 ticks
Avg Turnaround Time	109.00 ticks
Avg Response Time	98.20 ticks
CPU Utilization	98.5%
Idle CPU Time	4 ticks (1.5%)
Throughput	0.0912 proc/tick
Context Switches	24
Fairness Index	0.636
Starvation Score	1.27

Scheduling Efficiency Score	27.6 / 100
Energy Score (simulated)	342 units
Cache Miss Rate (simulated)	6.0%

Detail — Round Robin



Execution timeline (Gantt) — Round Robin



Busy vs. idle CPU time — Round Robin

Metric	Value
Avg Waiting Time	133.40 ticks
Avg Turnaround Time	144.20 ticks
Avg Response Time	39.00 ticks
CPU Utilization	98.5%
Idle CPU Time	4 ticks (1.5%)
Throughput	0.0912 proc/tick
Context Switches	73
Fairness Index	0.783
Starvation Score	0.63

Scheduling Efficiency Score	47.2 / 100
Energy Score (simulated)	489 units
Cache Miss Rate (simulated)	84.5%